Exact Polynomial Budgets for Product-Projection Localization

A Scoped Ram Analysis of the QAC^0 Parity Manuscript

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Abstract

The assigned manuscript claims a parity lower bound for fixed-depth circuits with polynomially many qubits, initially zero ancillas, one measured output, and unrestricted discarded registers. We analyze its discrete-count polynomial module and its ordered operator interface, without claiming independent certification of the complete circuit theorem. Positive roots give an exact weighted coefficient norm, retaining the normalization that a triangle estimate discards. The maximum error on the integer grid is attained at its right endpoint. Moving the Chebyshev interval to the first unannihilated integer gives an exact finite degree rule, and choosing a smaller root block reduces the displayed construction's logarithmic degree overhead. We derive residual certificates, a Ram moment realization with the correct coefficient metric, and a finite transition inequality retaining the inverse-circuit tail. These statements do not turn discrete approximation into interval approximation or remove the source's depth and scale inductions. Classical Chebyshev interpolation, finite moment unisolvence, and Bernstein enclosures are explicitly credited. Contribution assessment: C2/5, exact finite budgets and a logarithmic degree refinement of a classical source construction, without a new circuit lower bound or historical approximation claim.

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1 The actual problem and its proof dependencies

All logarithms below are natural unless specified. The report concerns the captured September 24 manuscript [1], not a live-branch assertion. Its eleven cached text files were freshly read in full. The source's declared result is not treated as an accepted external theorem merely because its proof is present. We reproduce and refine selected finite steps; its complete uniform circuit conclusion remains an upstream claim in this report.

1.1 Registers, projections, and norms

The physical space is $\mathcal{H}_N = (\mathbb{C}^2)^{\otimes N}$, with its usual Hilbert norm and induced operator norm. There are n classical input bits and $N - n$ ancillas, initialized as $|x\rangle \otimes |0^{N-n}\rangle$. Only a designated output qubit is measured in the computational basis. The other registers may finish in arbitrary, input-dependent states. They need not preserve the input or return the ancillas to zero. Intermediate measurements, postselection, and primitive unbounded fanout are not allowed.

The gates are arbitrary single-qubit unitaries and unbounded-arity Toffoli gates, with disjoint supports within each layer. The source enlarges the description to local product unitaries alternating with disjoint reflections $I - 2A$, where A is a product projection. A Toffoli is of this form with a computational-basis control projection and target $|-\rangle\langle -|$. Reflection depth is bounded by the original circuit depth; inversion reverses the layers and stays in the enlarged class.

For a subset $S \subseteq [N]$ and unit vectors v_i, set

$$M = \sum_{i \in S} (I - |v_i\rangle\langle v_i|)_i. \tag{1}$$

Its commuting summands have spectrum $\{0, 1\}$, so $\operatorname{spec} M \subseteq \{0, \ldots, |S|\}$. The projection $[M = 0]$ is the identity on uncounted registers, not necessarily a rank-one projection on all of $\mathcal{H}_N$. A second count D need not commute with M, even though each count has commuting internal summands.

The source localization assertion $\mathcal{B}_d(s, t)$, for fixed depth d and $0 \le s < t$, is

$$\left\| [M \ge N^t] U [D = 0] U^* [M = 0] \right\| \le e^{-N^s} \tag{2}$$

for all sufficiently large N, uniformly in both counts and all allowed depth-d circuits. Its threshold is not asserted uniform in every real exponent pair simultaneously. When $t > 1$, the high projection vanishes.

1.2 Architecture and retained details

Source module	Mechanism and boundary retained here
`localization`	Depth-zero estimate includes the amplitude lost on the initial projection, not only a normalized conditional Bernoulli law.
`depth_reduction`	Orthogonal mismatch sectors, small-support reflection expansion, and three ordered projection factors.
`polynomial`	Roots at low integers, Chebyshev damping on a remaining interval, degree and weighted coefficient budgets.
`bootstrap`	Replace only the first projection after squaring; the remaining tail is controlled using the inverse circuit at a larger scale. A finite backward chain closes the depth induction.
`parity`	Complementary Hadamard-product matrix entries average to the measured-output parity coefficient. Garbage is unrestricted.
`consequences`	State-preparation obstruction, scalar Fourier concentration, and symmetric-function consequences; some reductions and majority tail estimates are separately cited external inputs.

These files, the introduction, preamble, main file, bibliography, and README are all read, not merely inventoried. The external proofs cited by the consequences module were not all audited. No Lean execution or foreign checker replay is claimed.

For example, the depth-zero argument gives, for integer $r \geq 1$,

$$\|[M \geq r]Q[M = 0]\|^2 \leq \prod_i (1 - p_i) \Pr\{K \geq r\} \leq 2^{-r} \prod_i (1 - p_i)(1 + p_i) \leq 2^{-r},$$

where K is a sum of independent Bernoulli variables of parameters $p_i = 1 - |\langle w_i, v_i \rangle|^2$ on the overlapping supports. Indeed Markov's inequality for 2^K gives the middle step. The prefactor is essential; removing it can destroy the bound. Uncounted factors remain contractions even on entangled vectors. This correct source detail is not a new contribution.

2 The polynomial as an exact finite certificate

For a real polynomial $p(x) = \sum_{k=0}^{d} c_k x^k$, define

$$\mathcal{W}_N(p) = \sum_{k=0}^{d} |c_k| N^k, \qquad N > 0. \tag{3}$$

It is a coefficient norm, not the Hilbert norm of a state. It bounds $\|p(C)\|$ when $\|C\| \leq N$, and pays for the ordered monomials in a power of a count. We keep the grid endpoint m separate from the physical count bound N; in the source m can exceed N.

Let $T_0 = 1, T_1 = z, T_{j+1} = 2zT_j - T_{j-1}$ be the Chebyshev polynomials. Take integers $m \geq 2$, $0 \leq r \leq m - 2$, and $\ell \geq 0$. Choose $0 < L \leq r + 1$ with $L < m$, and put

$$\begin{aligned} F_r(x) &= \prod_{j=1}^{r} (1 - x/j), \qquad z_L(x) = \frac{m + L - 2x}{m - L}, \qquad z_0 = z_L(0) > 1, \\ p_{r,\ell,L}(x) &= F_r(x) \frac{T_\ell(z_L(x))}{T_\ell(z_0)}, \qquad B_r = \binom{m-1}{r}. \end{aligned} \tag{4}$$

Empty products are one. Two choices will matter: $L = r$ for $r \geq 1$ recovers the source's interval; $L = r + 1$ uses the first integer not already annihilated. The latter will be the default after the next theorem.

Theorem 2.1 (Exact grid error and weighted cost)**.** *For the parameters above,* $p(0) = 1$, $\deg p = r+\ell$, *and*

$$\max_{j\in\{1,\dots,m\}} |p(j)| = \frac{B_r}{T_\ell(z_0)}, \tag{5}$$

$$\mathcal{W}_N(p) = p(-N) = \left(\prod_{j=1}^{r}(1+N/j)\right)\frac{T_\ell((m+L+2N)/(m-L))}{T_\ell(z_0)}. \tag{6}$$

For integer $N \geq 1$*, the product in* (6) *is* $\binom{N+r}{r}$*. Both identities allow repeated roots.*

Proof. The positive zeros of F_r annihilate the first r integers. For $x \in [r+1, m]$, each $x - j$ is positive and increasing, hence

$$|F_r(x)| = \frac{\prod_{j=1}^{r}(x-j)}{r!} \leq B_r.$$

This interval is contained in $[L, m]$, where $|T_\ell(z_L(x))| \leq 1$. At $x = m$, the first bound is equality and $T_\ell(-1) = (-1)^\ell$, so the grid maximum is exactly (5). Normalization follows at zero, and $T_\ell(z_0) = \cosh(\ell \operatorname{arcosh} z_0) > 0$.

For $\ell \geq 1$, the zeros of T_ℓ are $\cos((2j-1)\pi/(2\ell))$, $1 \leq j \leq \ell$. Their inverse images under z_L lie in (L, m), and hence are strictly positive. Thus $p(x) = \prod_{j=1}^{r+\ell}(1 - x/\rho_j)$, where all $\rho_j > 0$. Its coefficient of x^k is $(-1)^k$ times a positive elementary symmetric function of the reciprocal roots. Consequently $\mathcal{W}_N(p) = \sum_k (-1)^k c_k N^k = p(-N)$. This argument neither requires disjoint root sets nor simple roots. The degree and formula follow; $\ell = 0$ is the same empty-factor case. □

Corollary 2.2 (Exact degree choice and interval shift)**.** *For fixed* m, r, L *and target* $0 < \delta < 1$*, the smallest integer* $\ell \geq 0$ *in this family with grid error at most* δ *is*

$$\ell_* = \left\lceil \frac{\operatorname{arcosh}(B_r/\delta)}{\operatorname{arcosh}((m+L)/(m-L))} \right\rceil. \tag{7}$$

Within $0 < L \leq r+1$*, taking* $L = r+1$ *minimizes this required degree. For rational* δ, L*, its correctness and minimality can instead be certified by exact rational recurrence comparisons* $T_{\ell_*}(z_0) \geq B_r/\delta > T_{\ell_*-1}(z_0)$.

Proof. The right side of (5) is $B_r/\cosh(\ell\alpha_0)$, where $\alpha_0 = \operatorname{arcosh} z_0 > 0$. It decreases strictly with integer $\ell \geq 0$. Moreover $(m+L)/(m-L)$ increases strictly with L. These observations give the formula and the permitted interval improvement. Here $B_r/\delta > 1$, so $\ell_* \geq 1$ and the preceding-index check is meaningful. □

For the source choice $L = r$, $|F_r|$ is also monotone on $[r, m]$ when $r \geq 1$, with maximum B_r. Thus its existing polynomial already has the exact costs above; these identities do not diagnose a source error. The displayed source estimates m^r and a recurrence majorant are valid but deliberately loose. Moving the interval to $r+1$ does not claim small values between the annihilated integers.

The omitted endpoint choices have elementary alternatives. If all m positive grid points are annihilated, use F_m, with degree m, zero grid error, and $\mathcal{W}_N(F_m) = \binom{N+m}{m}$ for integer N. At $r = m-1$, the shifted interval degenerates to a point and (4) must not be used. A finite search can compare F_m against every admissible r and its exact ℓ_*; this is a search over a specified family, not global optimal polynomial approximation.

3 A smaller root block and its actual asymptotics

The source takes $E = N^a$, $r = \lceil E \rceil$, $m = \lceil N^b \rceil$, and $\ell = \lceil 2\sqrt{mr}\log m \rceil$, for fixed $0 < a < b$. Its degree is $O(N^{(a+b)/2}\log N)$, and it proves $\log \mathcal{W}_N(p) = O(N^{(a+b)/2}(\log N)^2)$. These suffice for every fixed exponent strictly above $(a+b)/2$. The following choice improves those overheads while keeping that strict exponent condition unchanged.

Theorem 3.1 (Root allocation for the source regime)**.** *Fix $0 < a < b$, set $E = N^a$, $m = \lceil N^b \rceil$, and let*

$$H = \log(m/E), \quad r = \lceil E/H \rceil, \quad L = r + 1, \quad \ell = \left\lceil \frac{E + \log(2B_r)}{\alpha_0} \right\rceil, \quad \alpha_0 = \operatorname{arcosh} \frac{m+L}{m-L}. \tag{8}$$

For all sufficiently large integer N, these are admissible and the polynomial (4) *satisfies*

$$\max_{1 \le j \le m} |p(j)| \le e^{-E}, \tag{9}$$

$$\deg p \le (1 + o(1))\sqrt{mE\log(m/E)}, \tag{10}$$

$$\log \mathcal{W}_N(p) = O\Big(N^{(a+b)/2}(\log N)^{3/2}\Big). \tag{11}$$

Constants and the $o(1)$ depend on the fixed pair (a, b). In particular, for every fixed $u > (a+b)/2$, eventually $\deg p \le N^u$ and $\mathcal{W}_N(p) \le e^{N^u}$.

Proof. We have $H = (b-a)\log N + o(1) \to \infty$ and $E/H \to \infty$. Hence $r = (1 + o(1))E/H$, $L = (1 + o(1))E/H$, and $L/m \to 0$, so $1 \le r \le m - 2$ eventually. The identity

$$\alpha_0 = 2\operatorname{artanh}\sqrt{L/m} \ge 2\sqrt{L/m}$$

follows by evaluating the hyperbolic cosine, or its logarithmic formula. Also $\cosh v \ge e^v/2$, so the selected ℓ gives $T_\ell(z_0) \ge e^E B_r$, proving (9).

The elementary factorial inequality $r! \ge (r/e)^r$ gives

$$\log B_r \le r \log \frac{e(m-1)}{r} \le r\big(1 + \log(m/r)\big) = (1 + o(1))E,$$

because $\log(m/r) = H + \log H + o(1)$. It follows that

$$\ell \le 1 + \frac{(2 + o(1))E}{2\sqrt{L/m}} = (1 + o(1))\sqrt{mEH}.$$

The ratio $r/\sqrt{mEH}$ tends to zero, proving (10). This is an upper bound for a construction, not a matching lower bound for all polynomial families.

For the coefficient budget, Theorem 2.1 gives

$$\log \mathcal{W}_N(p) \le r\log(1+N) + \ell\alpha_N, \qquad \alpha_N = \operatorname{arcosh}\frac{m + L + 2N}{m - L},$$

where we have used $\cosh(\ell\alpha_N) \le e^{\ell\alpha_N}$ and only for this asymptotic estimate dropped a denominator at least one. Since $L/m \to 0$, the argument of arcosh is bounded by a constant times $1 + N$, whence $\alpha_N = O(\log N)$. Combining this with $\ell = O(N^{(a+b)/2}\sqrt{\log N})$ and $r = O(N^a/\log N)$ proves (11). Any fixed positive power gap absorbs the logarithms, proving the final assertion. No uniform threshold as $b - a \downarrow 0$ is asserted. □

The selection of an interpolation block of order $\log(1/\delta)/\log m$ is classical in spirit and is already present in Sherstov's Lemma 3.9 [2]. The result above is its explicit one-sided allocation within this source's positive-root family, with the weighted coefficient budget needed here. It is not a historically new approximate-degree bound. More elaborate classical approximants can have better unweighted degree than this particular family; their weighted coefficient cost and the ordered circuit interface would still need proof.

3.1 The degree–coefficient tradeoff is genuinely two-dimensional

Write $A_N = \operatorname{arcosh} z_L(-N)$. The exact logarithmic coefficient budget is

$$\log \mathcal{W}_N(p) = \sum_{j=1}^{r} \log(1 + N/j) + \log\cosh(\ell A_N) - \log\cosh(\ell\alpha_0). \tag{12}$$

For fixed r, L, this increases strictly with real $\ell > 0$: its derivative is $A_N \tanh(\ell A_N) - \alpha_0 \tanh(\ell\alpha_0) > 0$, since $A_N > \alpha_0$ and $v \mapsto v\tanh(\ell v)$ is increasing. Thus the smallest admissible ℓ also minimizes this family's cost for that root block. Different r can trade degree against cost; there is no assertion that minimizing one minimizes the other.

For evaluation without subtracting two enormous hyperbolic cosines, use

$$\log\cosh v = v - \log 2 + \log(1 + e^{-2v}), \qquad v \geq 0. \tag{13}$$

The last term lies in $[0, \log 2]$. Verified interval implementations can use this identity, but ordinary floating-point output alone is not a certificate. Exact rational recurrence provides the finite comparisons in Corollary 2.2; its integer bit sizes are a cost, not hidden constant-time operations.

4 Evaluation, perturbations, and finite Ram data

4.1 A recurrence-residual certificate on the correct interval

The identities above remove large pessimistic constants but do not excuse unverified numerical cancellation. The following elementary recurrence bound permits an independent check of an evaluated numerator.

Proposition 4.1 (Chebyshev residual budget)**.** *Fix $z \in [-1, 1]$, integer $\ell \geq 1$, and real numbers $\widehat{t}_0, \ldots, \widehat{t}_\ell$. Set*

$$e_0 = \widehat{t}_0 - 1, \quad e_1 = \widehat{t}_1 - z, \quad r_j = \widehat{t}_{j+1} - 2z\widehat{t}_j + \widehat{t}_{j-1} \quad (1 \leq j < \ell).$$

Then

$$|\widehat{t}_\ell - T_\ell(z)| \leq R_\ell := \ell|e_1| + (\ell - 1)|e_0| + \sum_{j=1}^{\ell-1} (\ell - j)|r_j|. \tag{14}$$

For $x \in [r + 1, m]$, suppose $\widehat{F} \in \mathbb{R}$, $\widehat{d} > 0$, and

$$|\widehat{F} - F_r(x)| \leq \eta_F, \quad |\widehat{d} - T_\ell(z_0)| \leq \eta_d < \widehat{d}.$$

Using $z = z_L(x)$, the evaluated quotient obeys

$$\left| \frac{\widehat{F}\widehat{t}_\ell}{\widehat{d}} - p(x) \right| \leq \frac{|\widehat{F}|R_\ell + \eta_F}{\widehat{d}} + \frac{B_r\eta_d}{\widehat{d}(\widehat{d} - \eta_d)}. \tag{15}$$

Proof. Let U_j be the second-kind Chebyshev polynomials, with $U_{-1} = 0$. Solving the recurrence for the error gives

$$e_\ell = U_{\ell-1}(z)e_1 - U_{\ell-2}(z)e_0 + \sum_{j=1}^{\ell-1} U_{\ell-1-j}(z)r_j.$$

For $z = \cos\theta$, the finite geometric sum representation of $\sin((j+1)\theta)/\sin\theta$ gives $|U_j(z)| \le j+1$, including endpoints by continuity. This proves (14). For the quotient, insert first $\widehat{F}T_\ell(z)/\widehat{d}$, then $F_r(x)T_\ell(z)/\widehat{d}$, and then the exact quotient. Use $|T_\ell(z)| \le 1$, $|F_r(x)| \le B_r$, and $T_\ell(z_0) \ge \widehat{d} - \eta_d > 0$. □

For $z \notin [-1,1]$ the polynomial bound on U_j is unavailable: the hyperbolic growth must be retained. Thus the proposition certifies the numerator on its actual interval, not the normalizer at $z_0 > 1$. The latter can be computed exactly for rational parameters, or enclosed separately. A floating-point recurrence without bounded residuals and a denominator margin does not acquire a stability theorem from this formula. The first r integer evaluations should be represented by their exact zero factors, rather than inferred from cancellation in the expanded basis.

4.2 A concrete finite moment realization

The freshly read original TR250 [7] specifies physical spaces before coordinate matrices and proves unisolvence of positive-weight polynomial moments. Here a particularly transparent bank represents $g(s) = p(Ns) = \sum_{k=0}^{d} a_k s^k$ on $[0,1]$:

$$b_j = \int_0^1 s^j g(s)\,ds, \qquad E_{jk} = \frac{1}{j+k+1}, \qquad b = Ea, \quad 0 \le j,k \le d. \tag{16}$$

For every nonzero real a, $a^T Ea = \int_0^1 (\sum a_k s^k)^2 ds > 0$, so E is invertible. Consequently the circuit-relevant budget in these coordinates is exactly

$$\mathcal{W}_N(p) = \left\| E^{-1} b \right\|_1, \tag{17}$$

not $\|b\|_2$. If measured moments obey $|\widehat{b}_j - b_j| \le \epsilon_j$, then

$$\mathcal{W}_N(\widehat{p} - p) \le \sum_{j=0}^{d} \epsilon_j \sum_{k=0}^{d} |(E^{-1})_{kj}|. \tag{18}$$

This follows directly by applying the triangle inequality to $\widehat{a} - a = E^{-1}(\widehat{b} - b)$; no conditioning benefit is claimed. Indeed Hilbert-type moment matrices can be poorly conditioned.

When the positive-root structure of the *exact* polynomial is known, Theorem 2.1 also writes the scalar cost as $v^T E^{-1} b$, where $v_k = (-1)^k$. For an arbitrary perturbation this scalar alone is not a norm: the error polynomial $1 + s$ vanishes at $s = -1$ while its coefficient norm is two. Thus a noisy moment reconstruction must use (18), or must independently certify the required alternating signs. An invertible representation neither improves polynomial degree nor makes noisy data exact.

The original local-moment chapter [9] proves a Taylor formula with integral remainder, exact on a fixed polynomial space, and explicitly distinguishes exactness from degree closure. That is the appropriate analogy here: (16) is an exact finite representation, whereas replacing a nonpolynomial function by a finite jet would need its own remainder and domain proof. No differential approximation to a quantum gate is being made. The chapter's included companion files were not read in this assignment; the theorem bodies used here occur in the chapter file itself.

4.3 Grid, interval, and coefficient uncertainty are different

The Bernstein hull and exact subdivision proofs in the original clearance chapter [8] give a way to check real-interval claims when such claims are actually needed. For a polynomial $v(s) = \sum_{k=0}^{d} \alpha_k s^k$, its degree-$d$ Bernstein coefficients are $\beta_j = \sum_{k \le j} \alpha_k \binom{j}{k} / \binom{d}{k}$. Since the Bernstein basis is nonnegative and sums to one, $\min \beta_j \le v(s) \le \max \beta_j$ on $[0,1]$. The conversion follows by expanding $s^k \sum_{j=k}^{d} \binom{d-k}{j-k} s^{j-k}(1-s)^{d-j} = s^k$. This classical certificate can bound the numerator between grid points, but it is not needed to justify a count whose spectrum is exactly integral.

Example 4.2 (A grid certificate need not be an interval certificate)**.** Take $m = 8$ and the exact annihilator F_8. It is zero at every integer $1, \dots, 8$, yet $F_8(3/2) = -\binom{16}{8}/(15 \cdot 4^8) \neq 0$. It cannot satisfy a zero-error bound on the whole interval $[1, 8]$. More generally, no polynomial with value one at zero can vanish on an interval. Positive-root structure does not remove this distinction.

If a coefficient perturbation is bounded in $\mathcal{W}_N$, it controls $\|\widehat{p}(C) - p(C)\|$ for $\|C\| \le N$ and scalar values on $[-N, N]$. When $m > N$, it does *not* automatically control all of $\{1, \dots, m\}$; that needs $\mathcal{W}_m$ or a separate grid bound. Likewise, replacing the exact integer spectrum by a nearly integral spectrum requires derivative or interval estimates near each integer. Tiny root displacement is not the same as exact annihilation.

5 The ordered finite operator interface

We next give a finite statement that can use the exact budgets without silently importing a global localization theorem. Let $D = \sum_{i=1}^{q} P_i$, with $q \ge 1$, where the P_i are commuting one-qubit projections on distinct sites, and let U be unitary on $\mathcal{H}_N$. Set $C = UDU^*$, $Q = [C = 0]$. Let P, H be orthogonal projections with $HP = 0$. In the source these are $[M = 0]$ and $[M \ge N^t]$. They are not assumed to commute with C. For the omitted empty count, $D = 0$, $Q = I$, and $HQP = HP = 0$ is immediate without defining $\mathcal{W}_0$.

For a polynomial p, write $\Delta^j p(0) = \sum_{i=0}^{j} (-1)^{j-i} \binom{j}{i} p(i)$. Suppose that for every nonempty $S \subseteq [q]$, $|S| = j$,

$$\left\| HU\Big(\prod_{i \in S} P_i\Big)U^*P \right\| \le \kappa_j. \tag{19}$$

These are actual operator bounds, not entries in a sampled matrix.

Theorem 5.1 (Finite transition with exact grid and support costs)**.** *Let $p(0) = 1$, $\deg p = d$, and let $m \ge 1$ be an integer. Define*

$$\begin{aligned} \delta_p &= \max_{1 \le i \le \min(m,q)} |p(i)|, \\ G_p &= \max_{m < i \le q} |p(i)|, \qquad \tau = \|[C > m]PQ\|, \\ V_\kappa(p) &= \sum_{j=1}^{\min(d,q)} \binom{q}{j} |\Delta^j p(0)| \kappa_j. \end{aligned}$$

An empty maximum or sum is zero. Under (19)*,*

$$\|HQP\|^2 \le V_\kappa(p) + \delta_p + G_p \tau. \tag{20}$$

If $\kappa_j \le \kappa$ throughout, then

$$V_\kappa(p) \le \kappa V_q(p), \qquad V_q(p) := \sum_{j=1}^{\min(d,q)} \binom{q}{j} |\Delta^j p(0)| \le \mathcal{W}_q(p) - 1. \tag{21}$$

For the positive-root family, the last quantity is explicitly $p(-q) - 1$.

Proof. Newton interpolation on the integral spectrum gives

$$p(D) = \sum_{j=0}^{\min(d,q)} \Delta^j p(0) \binom{D}{j}, \qquad \binom{D}{j} = \sum_{|S|=j} \prod_{i \in S} P_i.$$

The second identity follows on a simultaneous eigenvector: if precisely k projections equal one, each side equals $\binom{k}{j}$. The first is the finite-difference expansion of a polynomial, restricted to this same spectrum. Its constant term contributes nothing because $HP = 0$. Thus $\|Hp(C)P\| \le V_\kappa(p)$.

Crucially, square the desired norm before substituting:

$$\begin{aligned} \|HQP\|^2 &= \|HQPQH\| \\ &\le \|Hp(C)PQH\| + \|H(Q - p(C))PQH\| \\ &\le V_\kappa(p) + \|(Q - p(C))PQ\|. \end{aligned}$$

At eigenvalue zero the last difference is zero. At positive eigenvalues at most m its modulus is at most δ_p. Above m, $Q = 0$, so its modulus is at most G_p, not necessarily $1 + \|p(C)\|$. Splitting at these spectral projections and using their commutation with $p(C)$ gives (20). No commutation of P with C has been invoked.

Finally, for $k \ge 1$, $\Delta^j x^k|_{x=0} = j!\, S(k,j)$, where $S(k,j) \ge 0$ counts partitions into j nonempty blocks. Counting maps from a k-element set into q labels by their image size gives $\sum_j \binom{q}{j} j! S(k,j) = q^k$. Apply these identities and the triangle inequality to $p = \sum c_k x^k$. The result is $V_q(p) \le \sum_{k \ge 1} |c_k| q^k = \mathcal{W}_q(p) - 1$. □

The Newton identity is classical; its use here merely collapses repeated indices before paying the actual support-dependent localization cost. It need not be sharp for every polynomial. The theorem permits exact finite differences, direct grid evaluation of G_p, and the positive-root coefficient formula to be compared without changing the physical operators. No estimate of κ_j or τ is generated by interpolation.

5.1 How it joins, and does not replace, the source induction

In the source, powers D^k are expanded into at most N^k ordered monomials. Uniform small-support localization supplies their bound, and the coefficient budget absorbs their number. Theorem 5.1 is a finite refinement of this step. Since $q \le N$, its conservative bound $V_q \le \mathcal{W}_N - 1$ never worsens that counting budget.

For a source backward step, fixed parameters must still satisfy

$$0 \le s < t < b < 2t - s, \qquad s < a < 2t - b, \qquad (a+b)/2 < u < w < t.$$

The source proves small-support bounds at depth d from localization at depth $d-1$, expanding at most 9^k absolute coefficient weight for k affected gates and retaining the three-factor order. It then uses the already established same-depth statement at a larger threshold for the inverse circuit:

$$\left\| [C \ge N^b] PQ \right\| = \left\| [D \ge N^b] U^* [M = 0] U [D = 0] \right\| \le e^{-N^t}.$$

The polynomial theorem above supplies the same asymptotic degree and coefficient exponent conditions, with smaller logarithmic overhead. It does not relax $u > (a+b)/2$, produce a threshold uniform in all real parameters, or replace that inverse hypothesis.

For completeness, the finite scale chain in the source is $s_{i+1} = t_i$, $t_{i+1} = 2t_i - s_i - 3\eta$, $\eta = (t-s)^2/100$. Writing $g = t - s$, its gaps are $g_i = g - 3i\eta$. For $K = \lceil 2/g \rceil$ with $0 < g \le 1$, $3K\eta \le 0.09g < g/2$, so $t_K > 1$. The induction runs *backward* from this trivial tail and uses finitely many fixed tuples at each fixed depth. Taking their maximum threshold is legitimate; claiming a single threshold for all exponent tuples would be a different assertion.

Lastly, for an actual circuit W, let $B = W^* O_0 W$ and $p_0(x) = \langle x, 0|B|x, 0\rangle$. With Hadamard product vectors $\xi_h = 2^{-n/2}\sum_x (-1)^{h\cdot x}|x\rangle$, character orthogonality gives

$$\mathbb{E}_h \langle \xi_{\bar{h}}, 0|B|\xi_h, 0\rangle = 2^{-n}\sum_x (-1)^{|x|} p_0(x).$$

Worst-case parity success $1/2 + \varepsilon$ makes this at least ε. The source localization claim bounds each entry by e^{-N^s}, choosing $0 < s < t < 1/c$ when $n \le N \le n^c$. This displays the measured-output interface and why arbitrary final garbage is allowed. It also displays precisely the global uniform input that a finite polynomial certificate cannot supply.

6 Worked finite examples and check scope

Example 6.1 (A fully rational budget)**.** Take $m = N = 8$, $r = 1$, $L = 2$, and target $\delta = 1/4$. Here $B_r = 7$, $z_0 = 5/3$, and

$$T_3(5/3) = 365/27 < 28, \qquad T_4(5/3) = 3281/81 > 28.$$

Thus $\ell_* = 4$, the total degree is five, and

$$\max_{1\le j\le 8} |p(j)| = \frac{567}{3281} < \frac{1}{4}, \qquad \mathcal{W}_8(p) = \frac{1947609}{3281}.$$

The cost is substantial despite the small grid error. One must not replace it by the grid error in the monomial expansion. For comparison, keeping $L = r = 1$ requires $\ell_* = 6$ for this same root block and target. This finite example is not an asymptotic benchmark or a circuit simulation.

Example 6.2 (Repeated roots and endpoint conventions)**.** With $r = 1, m = 3, L = 1, \ell = 1$, the roots are one and two. If another choice makes a Chebyshev inverse-image root coincide with one of $1, \dots, r$, the elementary-symmetric proof still applies with multiplicity. Such coincidence can occur when $L < r$; for example $r = 2, m = 3, L = 1, \ell = 1$ has the common root two, and is valid under the same positive-root reasoning though outside the default $r \le m - 2$ search convention. No denominator vanishes: all roots are strictly positive and the normalization point remains zero.

The owned checker uses exact rational polynomial arithmetic, independently expanding products and Chebyshev recurrences. It checks coefficient signs, the negative-point norm identity, the maximum over every integer in each finite grid, predecessor/minimal-degree comparisons, Newton support costs, moment reconstruction and perturbation bounds, and recurrence-residual identities including endpoint cases. Its record distinguishes parameter cases from individual assertions: 682 exact rational cases and 12,447 assertions passed, with zero numerical cases. It does not execute another author's checker, sample a purported circuit theorem, or turn a finite list into proof of the all-N statements. The proofs above establish those local statements; fixtures diagnose transcription and implementation errors.

7 Current context, costs, and remaining work

7.1 Literature and attribution as of October 9, 2026

The freshly opened primary record of Joshi–Tal–Vasconcelos–Wright [4] is arXiv v4, August 25, 2026, with the STOC 2026 publication listed. Its abstract distinguishes depth-three parity lower bounds from depth-two approximation bounds with one output bit. The primary records of Gretta–Gupta–Joshi [5] (v2, August 25) and Xu–Li [6] (v1, September 4) describe Fourier-concentration and fanout reductions; their abstract/version pages, not their complete proofs, were inspected. These are external context, not an independent endorsement of the assigned source's arbitrary fixed-depth assertion.

Sherstov's author manuscript [2], Section 3.1 and Lemma 3.9 with its proof, was inspected. Its interpolation and Chebyshev strategy already includes an accuracy-dependent root block. The construction here is compared to that classical method, not advertised as an improvement over the best known unweighted approximate degree. Anshu–Metger's publisher abstract [3] confirms the broader polynomial-localization context; its full proof was not audited here. The source's additional September preprint by Kintali was sought at its public URL, but the fetch failed; no fresh theorem verification is claimed for it. The publisher repository page for the assigned manuscript was accessible. Captured-tree byte identity is separate from publication review or live-branch status.

7.2 Advantages and costs

The exact negative-point formula retains normalization, avoids expanding all coefficients merely to total their absolute values, and remains valid with repeated positive roots. The shifted interval exploits the actual integer spectrum. Root allocation reduces the degree and coefficient overheads in the displayed source family. The finite operator theorem separates grid error, support localization, and the inverse tail rather than conflating them into one tolerance. These are mathematical benefits; no wall-clock speedup is measured.

The costs are equally explicit. Exact recurrences can generate large rational numerators and denominators. Searching root counts adds discrete optimization, and smaller degree need not minimize the weighted norm across different root blocks. A finite Ram moment matrix can worsen conditioning. Bernstein conversion can be conservative and is unnecessary for an exact integer spectrum. Perturbing the roots or the count spectrum forfeits exact zeros unless additional interval margins are paid. All circuit estimates must remain in the physical Hilbert operator norm, not a coordinate norm.

7.3 Plausible extensions and missing premises

One useful next step is a certified Pareto search over (r, ℓ, L) for fixed finite (N, m, δ), comparing degree with the exact $\mathcal{W}_N$ and Newton support costs. A general nonuniform annihilation set could exploit a known spectrum, but both its maximum root-factor envelope and its positive-root/coefficient structure would need reanalysis. Replacing the Chebyshev factor by an optimal discrete approximant may improve degree, but is not useful to this proof without a simultaneous coefficient budget.

For a near-integral selfadjoint count, local spectral enclosures plus derivative bounds could replace exact annihilation; one must also control the stability of the zero spectral projection and the support expansion. For noncommuting count summands, neither the integer spectrum nor the subset Newton identity survives automatically. Finally, making the source depth/scale thresholds

effective would require tracking all its preceding localization constants, not just optimizing this polynomial. None of these extensions is asserted to prove a stronger parity theorem here.

8 Evidence boundary and conclusion

The literal-read ledger covers all 1,596 lines of the eleven cached source texts, including 182 bibliography lines and the 17-line README. Fresh private reads cover TR250 lines 121–181, 231–272, and 455–477 (126 lines), the 299-line clearance chapter, and the 446-line local-moment chapter file. The latter's included companion files are explicitly unread. The private total is 871 displayed lines, not the complete Ram corpus. PRE hashes were persisted before these displays; final POST hashes and unread complements are supplied in the owned read ledger. Searches and catalogue metadata are locators, not additional proof reads.

The contribution is deliberately scored C2 independently of L3 applicability. Its exact finite cost identities and root allocation are useful refinements of the source's deliberately coarse polynomial estimates. Classical interpolation and representation facts receive no new-discovery credit. No substantive source error is alleged in the steps read: the integer-grid condition, normalization, operator order, inverse closure, and finite scale chain are all retained. The global QAC^0 claim and the external consequences are not independently certified by this report or its tests.

References

[1] OpenAI, *Product-projection localization and the QAC0 parity lower bound*, September 24, 2026, captured manuscript. Publisher repository.

[2] A. A. Sherstov, *Approximate inclusion-exclusion for arbitrary symmetric functions*, Computational Complexity 18 (2009), 219–247, Section 3.1 and Lemma 3.9. `https://web.cs.ucla.edu/~sherstov/pdf/incl-excl.pdf`.

[3] A. Anshu and T. Metger, *Concentration bounds for quantum states and limitations on the QAOA from polynomial approximations*, Quantum 7 (2023), 999. `https://doi.org/10.22331/q-2023-05-11-999`.

[4] M. R. Joshi, A. Tal, F. Vasconcelos, and J. Wright, *Improved Lower Bounds for QAC0*, STOC 2026, 2199–2209; `https://arxiv.org/abs/2512.14643v4`.

[5] L. Gretta, M. Gupta, and M. R. Joshi, *Parity $\notin$ QAC0 $\iff$ QAC0 is Fourier-Concentrated*, `https://arxiv.org/abs/2604.02793v2`.

[6] B. Xu and L. Li, *Fanout Complexity of Symmetric Boolean Functions in QAC0*, `https://arxiv.org/abs/2609.05153v1`.

[7] M. Subbarao, original private `TR-rev-1-old-tr250.tex`, finite physical spaces, compatible coordinate transfer, and polynomial moment construction, lines 121–181, 231–272, 455–477. Locally inspected; not publicly transmitted.

[8] M. Subbarao, RamCalBooks, volume 16, `ch05d-clearance.tex`, original chapter, lines 1–299. Locally inspected Bernstein conversion, hull, subdivision, and error budgets.

[9] M. Subbarao, RamCalBooks, volume 1 revised, `ch03-local-moments-rev2.tex`, original chapter file, lines 1–446. Local Taylor remainder, degree closure, and moment-domain qualifications; included companion files not inspected.